

NARAYANA
IIT - JEE ACADEMY - INDIA

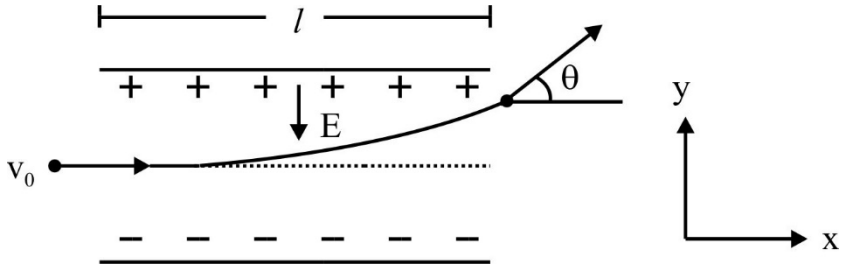
Sec: SR STAR - A & APEX, ALL IITS **DT: 19/01/2026**
Time: 3 Hrs **GTM - 19** **Max. Marks: 300**

PHYSICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

1. A uniform electric field $E = 91\times10^{-6}\ V/m$ is created between two parallel, charged plates as shown in figure. An electron enters the field symmetrically between the plates with a speed $v_0 = 4\times10^{3}\ m/s$. The length of each plate is $l = 1m$. Find the angle of deviation of the path of the electron as it comes out of the field. (Mass of the electron is $m = 9.1\times10^{-31}kg$ and its charge is $e = -1.6\times10^{-19}C$) (ignore gravity)



For More Join: @IITJEE_Advanc

A) $30°$ B) $45°$

C) $90°$ D) $60°$

2. **Assertion (A):** When identical cells are connected in parallel to an external load, the effective e.m.f increases.

Reason (R): All the cells will be sending unequal currents to the external load in the same direction.

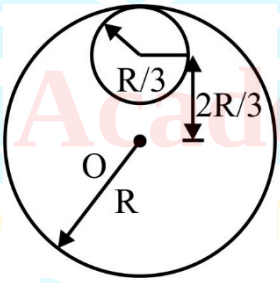
A) Both (A) and (R) are true and (R) is the correct explanation of (A)

B) Both (A) and (R) are true but (R) is not the correct explanation of (A)

C) (A) is true but (R) is false

D) (A) is false and (R) is false

3. A disc has mass $9m$ and radius R, a hole of radius $R/3$ is cut from it as shown in the figure. The moment of inertia of remaining part about an axis passing through the centre 'O' of the disc and perpendicular to the plane of the disc is_____



A) $\frac{37}{9}mR^2$ B) $\frac{40}{9}mR^2$ C) $4mR^2$ D) $8mR^2$

4. Heat required to convert $1g$ of ice at $0°C$ in to steam at $100°C$ is _____

(**Given:** Latent heat of Fusion is 80 Cal for 1 gram of ice and Latent heat of vaporization is 536 Cal for 1 gram of water. Specific heat capacity of water is $1\,cal/gm-°C$)

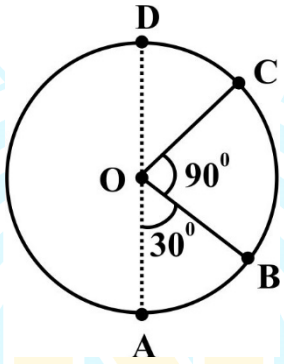
A) 100 Cal B) 200 Cal C) 700 Cal D) 716 Cal

5. What is the ratio of de-Broglie wavelength of electron in the second and third Bohr orbits in the hydrogen atom?

A) $\frac{2}{3}$ B) $\frac{3}{2}$ C) $\frac{4}{3}$ D) $\frac{3}{4}$

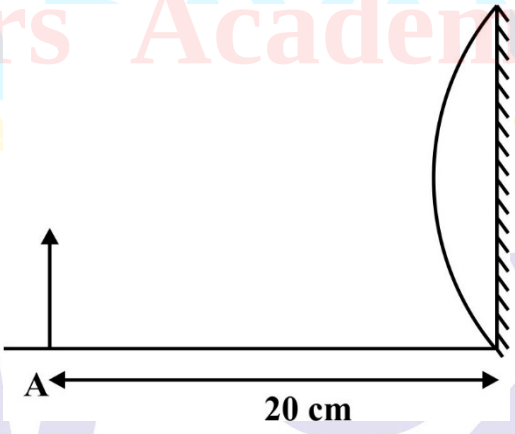
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6. A body of mass 100 g is moving in circular path of radius $2m$ on vertical plane as shown in figure. The velocity of the body at point A is $10\ m/s$. The ratio of its kinetic energies at point B and C is: (Take acceleration due to gravity as $10\ m/s^2$)



A) $\frac{2+\sqrt{3}}{3}$ B) $\frac{2+\sqrt{2}}{3}$ C) $\frac{3+\sqrt{3}}{2}$ D) $\frac{3-\sqrt{2}}{2}$

7. Focal length of the plano-convex lens is $15\ cm$. A small object is placed at A as shown in the figure. The plane surface is silvered. The image will form at

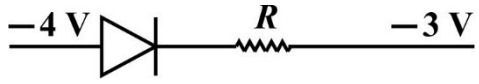


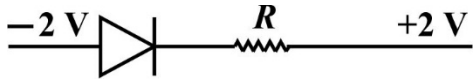
A) $60\,cm$ to the left of lens B) $12\,cm$ to the left of lens

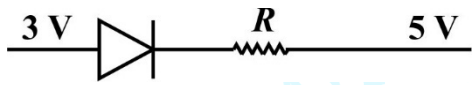
C) $60\,cm$ to the right of lens D) $30\,cm$ to the left of lens

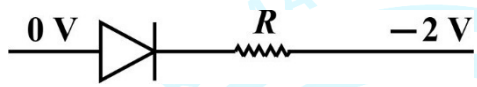
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8. Which one of the following represents forward bias diode?

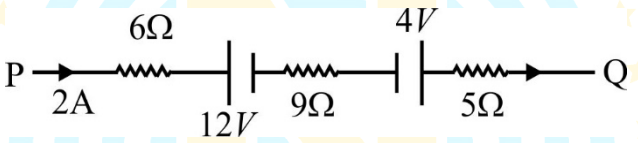
A) 

B) 

C) 

D) 

9. The potential difference between P and Q in the following figure is____



A) 24 V B) 14 V C) 32 V D) 48 V

10. A uniform solid sphere of mass 60 kg and radius 6m is kept fixed in space. A small point mass m is placed at a distance 18 m from the centre of the sphere. The gravitational force on m due to the complete sphere is F_1. Now, a spherical cavity of radius 2m is removed from the big sphere such that the centre of the cavity lies 4m from the big sphere's centre along the line joining the sphere and the point mass. After removing the cavity, the gravitational force on m, due to the remaining part is F_2, then the ratio of $F_1 : F_2$ is

A) $12:11$ B) $9:7$ C) $49:46$ D) $20:19$

11. The fundamental frequency of a closed organ pipe is same as the first overtone frequency of an open pipe. If the length of open pipe is $50\,cm$, then the length of closed pipe is

A) $25\,cm$ B) $12.5\,cm$ C) $100\,cm$ D) $200\,cm$

12. In SI units, the dimensions of $\sqrt{\frac{\epsilon_0}{\mu_0}}$ is:

A) $A^{-1}TML^3$ B) $AT^2M^{-1}L^{-1}$

C) $AT^{-3}ML^{3/2}$ D) $A^2T^3M^{-1}L^{-2}$

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13. A sphere of radius R and charge Q is placed inside a concentric imaginary sphere of radius $2R$. The flux associated with the imaginary sphere is

A) $\frac{Q}{\epsilon_0}$ B) $\frac{Q}{2\epsilon_0}$ C) $\frac{4Q}{\epsilon_0}$ D) $\frac{2Q}{\epsilon_0}$

14. In a Vernier calipers, each cm on the main scale is divided into 20 equal parts. 10 Vernier scale divisions are equivalent to 9 main scale divisions. The value of Vernier constant will be _____ $\times 10^{-2} mm$

A) 5 B) 50 C) 500 D) 200

15. Light of wavelength $550\ nm$ is incident on surfaces of cesium and lithium. Work functions of cesium and lithium are respectively $1.9\ eV$ and $2.5\ eV$. Then electron will be ejected from

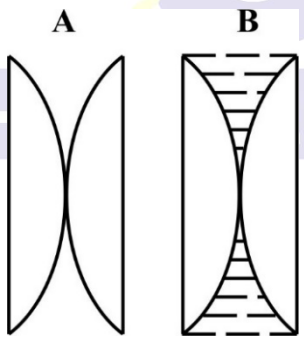
A) Cesium only B) Lithium only

C) From both of them D) None of them

16. Two spheres of the same material have radii $1\ m$ and $4\ m$ and temperatures $4000\ K$ and $2000\ K$ respectively. The ratio of the energy radiated per second by the first sphere to that by the second is

A) $1:1$ B) $16:1$ C) $4:1$ D) $1:9$

17. Figure A shows two identical plano-convex lenses in contact as shown. The combination has focal length $24\ cm$. Figure B shows the same with a liquid introduced between them. If refractive index of glass of the lenses is 1.50 and that of the liquid is 1.60, then the focal length of the system in figure B will be:



A) $-120cm$ B) $120cm$ C) $-24cm$ D) $24cm$

For More Join: @IITJEE_Advanc

18. **Assertion (A) :** If Young's double-slit experiment is performed in water instead of air, the separation between consecutive fringes decreases.

Reason (R): The wavelength of light reduces when it travels in a medium having refractive index greater than 1, while its frequency remains unchanged.

From the options below, choose the most appropriate answer.

A) Both (A) and (R) are true and (R) is the correct explanation of (A)

B) Both (A) and (R) are true but (R) is not the correct explanation of (A)

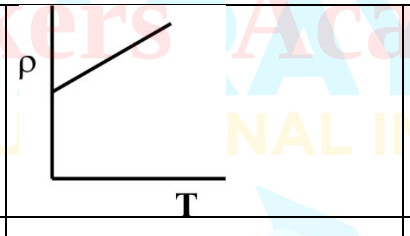
C) (A) is true but (R) is false

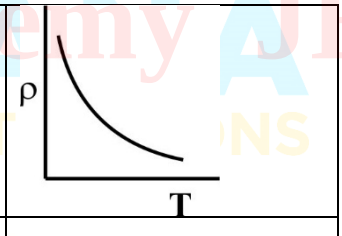
D) (A) is false but (R) is true

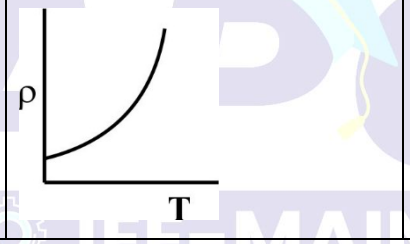
19. A capacitor stores $60\mu C$ charge when connected across a battery. When the gap between the plates is filled with a dielectric, a charge of $120\mu C$ flows through the battery. The dielectric constant of the material inserted is

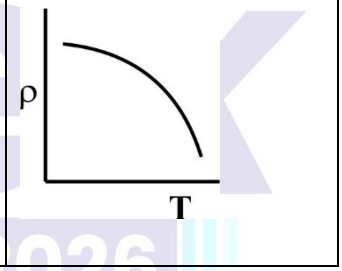
A) 1 B) 2 C) 3 D) 6

20. Which of the following graph represents the variation of resistivity (ρ) with temperature (T) for semiconductor?

A) 

B) 

C) 

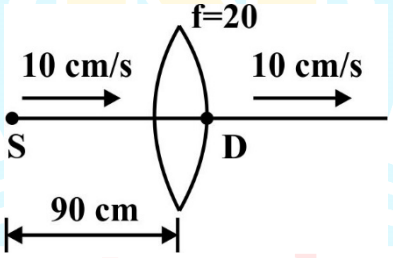
D) 

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SECTION – II
(Numerical Value Answer Type)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only**. Have to **answer 5 questions** and question will be evaluated according to the following marking scheme.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

21. The figure shows initial positions of a source of light S and a light detector D. Both S and D move with velocity $10\ cm/s$, as shown in the figure. The converging lens of focal length $20\ cm$ is fixed. If time instant(s) when the detector detects maximum intensity of light is t_1 and t_2 then, find $t_1 + t_2$ (in sec.)

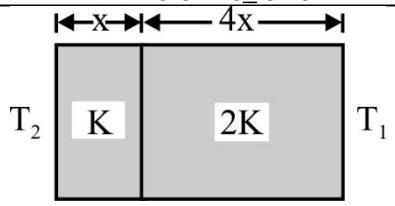


22. Two soap bubbles of radii $r_1 = 4cm$ and $r_2 = 20\ cm$ are in contact. What is the radius of curvature R (in cm) of the common surface?

23. A particle of mass 2 kg located at the position $\left(\hat{i} + \hat{j}\right)m$ has a velocity $2\left(\hat{i} - \hat{j} + \hat{k}\right)m/s$. Magnitude of angular momentum about z-axis in $kg - m^2/s$ is

24. The temperature of the two outer surfaces of a composite slab, consisting of two materials having coefficients of thermal conductivity K and $2K$ and thickness x and $4x$, respectively are T_2 and $T_1\left(T_2 > T_1\right)$. The rate of heat transfer through the slab, in a steady state is $\left(\frac{A\left(T_2 - T_1\right)K}{6x}\right)f$, then the value of f is_____ (A is the cross – sectional area of each slab)

For More Join: @IITJEE_Advanc



25. A body falls freely from rest. It covers as much distance in the last second of its motion as covered in the first three seconds. The body has fallen for a time of_____ (in sec.)

CHEMISTRY MAX.MARKS: 100

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

26. If the Bohr's radius of the hydrogen atom is 53 pm, the radius of the 1st orbit in He^+ ion is close to

A) 75 pm B) 38 pm C) 106 pm D) 27 pm

27. Which of the following molecule does not show geometrical isomerism?

A) But – 2 – ene B) 2, 3 – dichlorobut – 2 – ene

C) Pent – 2 – ene D) 2, 3 – dimethylbut – 2 – ene

28. For the process $H_2O_{(l)} \rightarrow H_2O_{(g)}$ at $T = 100^0C$ and 1 atm pressure, the correct choice is____

A) $\Delta S_{system} > 0$ and $\Delta S_{surroundings} > 0$

B) $\Delta S_{system} > 0$ and $\Delta S_{surroundings} < 0$

C) $\Delta S_{system} < 0$ and $\Delta S_{surroundings} > 0$

D) $\Delta S_{system} < 0$ and $\Delta S_{surroundings} < 0$

29. The outermost general electronic configuration of the group which contains most electronegative element is

A) ns^2np^3 B) ns^2np^4 C) ns^2np^5 D) ns^2np^6

For More Join: @IITJEE_Advanc

30. Arrange the following in decreasing order of ionic size.

A) $Ce^{3+} > Yb^{3+} > Gd^{3+} > Tb^{3+}$ B) $Ce^{3+} > Eu^{3+} > Tb^{3+} > Yb^{3+}$

C) $Eu^{3+} > Ce^{3+} > Tb^{3+} > Yb^{3+}$ D) $Tb^{3+} > Eu^{3+} > Ce^{3+} > Yb^{3+}$

31. Given below are two statements:

Statement (I): 4g of propyne reacts with excess of sodium to liberate half mole of H_2 gas.

Statement (II): One mole of propyne reacts with excess CH_3MgBr to liberate 22.7L of CH_4 at STP.

In the light of the above statements, choose the most appropriate answer from the options given below:

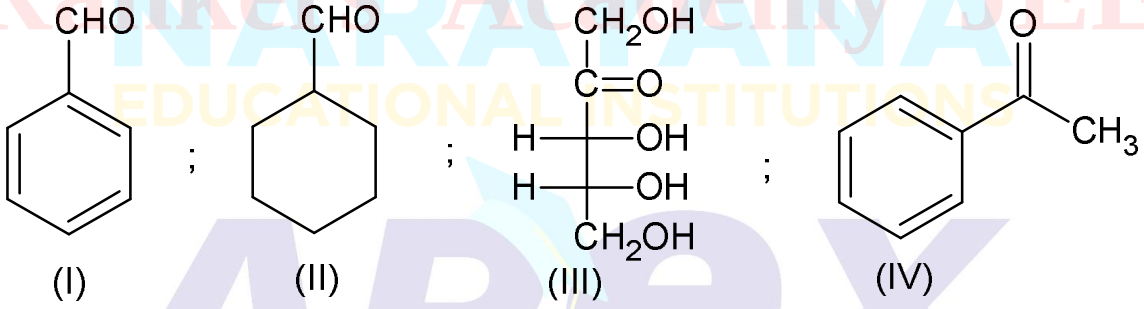
A) Statement I is false but Statement II is true

B) Both Statement I and Statement II are true

C) Statement I is true but Statement II is false

D) Both Statement I and Statement II are false

32. The compounds which can give positive Fehling's test are?



(I) ; (II) ; (III) ; (IV)

A) (II) and (III) only

B) (II), (III) and (IV) only

C) (I), (II) and (IV) only

D) (I), (III) and (IV) only

33. Which of the following aqueous solutions on electrolysis using inert electrodes liberate two different gases at anode.

A) $Mg(NO_3)_2$ B) KBr C) CH_3COONa D) $AgNO_3$

For More Join: @IITJEE_Advanc

34. **Statement-1:** The compound CH_2=CH-CH_2-Cl will undergo S_N1 reaction though it is a primary halide.

Statement-2: The compound CH_2=CH-Cl will not undergo S_N2 reaction as it is a vinyl halide.

In the light of the above statements, choose the most appropriate answer from the options given below:

A) Statement I is false but Statement II is true

B) Both Statement I and Statement II are true

C) Statement I is true but Statement II is false

D) Both Statement I and Statement II are false

35. Which of the following compounds is a vitamin?

A) Citric acid B) Tartaric acid C) Threonine D) Thiamine

36. Match list-I with list-II

List-I		**List-II**	
1)	C < Si < Ge < Sn < Pb	p)	Electronegativity
2)	B > Ga > T*l* > A*l* > In	q)	Melting point
3)	C > Si > Ge > Sn < Pb	r)	Ionic radius
4)	B > T*l* > In > Ga > A*l*	s)	3rd Ionisation potential

A) 1)-q : 2)-r : 3)-s : 4)-p B) 1)-r : 2)-s : 3)-q : 4)-p

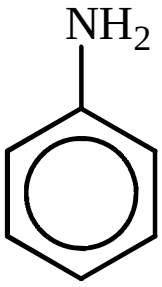
C) 1)-p : 2)-q : 3)-s : 4)-r D) 1)-q : 2)-p : 3)-s : 4)-r

37. A substance decomposes with first order with initial concentration of A (g/Lt). After time 't' sec it's concentration becomes $\frac{A}{n}(g/Lt)$. The rate constant (sec^{-1}) is

A) $\frac{\ln n}{t}$ B) $\frac{\ln \frac{1}{n}}{t}$ C) $\frac{\ln n}{t^2}$ D) $\frac{1}{nt}$

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38.



$\xrightarrow[\text{(ii) CuBr / HBr}]{\text{(i) } NaNO_2 \text{ / HCl, 273K}}$ Major product 'A'

(iii) Na - dry ether

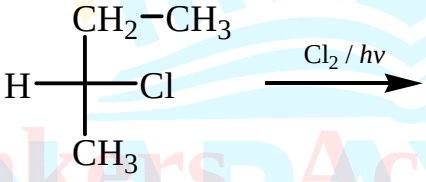
A) 'A' Contains 6 sp^2 carbon atoms

B) 'A' contains 6π bonds

C) Step – (iii) reaction is Wurtz reaction

D) GMW of A is 156

39. Total number of isomers possible (including stereo isomers) for the product in the following reaction is.



$(2R-2-\text{Chloro butane})$

A) 5 B) 6 C) 4 D) 8

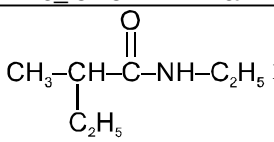
40. $A_{(g)} + 3B_{(g)} \rightleftharpoons 4C$ initial concentration of A is equal to that of B. The equilibrium concentrations of A and C are equal. The equilibrium constant K_C for the reverse of the above reaction is_____

A) 8 B) 0.08 C) 0.125 D) 1.25

41. On passing $'x'$ farads of electricity through an aqueous solution of K_2SO_4 using inert electrodes, 1.68 L (STP) of gases was obtained. How many moles of MnO_4^- could be reduced to Mn^{2+} by passing $'x'$ farads of electricity?

A) 0.10 B) 0.20 C) 0.15 D) 0.02

For More Join: @IITJEE_Advanc

42. The IUPAC name of the compound  is:

A) N-Ethyl-2-ethylpropanamide

B) N-Ethyl-2-methylbutanamide

C) N-Ethyl-2-methylpropanecarboxamide

D) N-Ethyl-2-ethylethanecarboxamide

43. Which of the following is correct statement

1) The metal – carbon bond in metal carbonyls possess both σ and π character

2) The $M-C$ σ bond is formed by the donation of lone pair of electrons on the carbonyl carbon into vacant orbital of the metal

3) The $M-C$ π bond is formed by the donation of a pair of electrons from a filled d – orbital of metal into the vacant anti bonding π^* orbital of carbon monoxide

4) The metal to ligand bonding in metal carbonyls creates a synergic – effect which strengthens the bond between carbon monoxide and the metal

A) 1, 4 B) 1, 2 C) 3, 4 D) All

44. We have four solutions of CH_3COOH (weak acid) labelled as A, B, C and D with concentrations 0.1M, 0.01M, 0.001M and 0.0001M respectively. The value of van't Hoff's factor for the solutions will be in the order.

A) $i_A < i_B < i_C < i_D$ B) $i_A > i_B > i_C > i_D$

C) $i_A = i_B = i_C = i_D$ D) $i_A < i_B < i_D < i_C$

45. The geometries of the complexes; $[Ni(CO)_4]$, $[Ni(CN)_4]^{2-}$ and $[NiCl_4]^{2-}$ respectively are

A) square planar, square planar, tetrahedral

B) tetrahedral, tetrahedral, square planar

C) tetrahedral, square planar, tetrahedral

D) tetrahedral, square planar, square planar

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46. The number of molecules/ions in which central atom is sp^3d hybridised is?

XeO_2F_2 , $XeOF_2$, XeF_2, ClF_3, PCl_4^-, BrF_5, $SnCl_2$, XeO_3, $XeOF_4$

47. The catalyst decrease the Ea from 100 KJ $mole^{-1}$ to 80 kJ $mole^{-1}$. The rate of catalysed reaction at $x\times10^2$K is equal to the rate of uncatalysed reaction at 5×10^2K . Find the value of 'x'?

48. The molecular weight of the compound (Q) in the following sequence of reactions is:

(Given: Atomic weight C=12, H=1, N=14)

$$\text{Benzyl amine} \xrightarrow[KOH]{CHCl_3} (P) \xrightarrow{LiAlH_4} (Q)$$

49. 25 ml of H_2O_2 solution was added to the excess of acidified KI solution. The iodine so liberated required 40 ml of 0.1 N sodium thiosulphate solution. If the normality of H_2O_2 solution is x , then the value $100x$ is

50. On heating 0.2 gm of an organic compound with a mixture of barium chloride and nitric acid, 0.466 gm of barium sulphate was obtained calculate the percentage of sulphur. (Given: Atomic weights of $Ba=137, S=32, O=16, H=1$)

For More Join: @IITJEE_Advanc

MATHEMATICS **MAX.MARKS: 100**

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51. Let $a_1, a_2, a_3, \ldots$ be an increasing G.P. of positive numbers. If $a_3 a_5 = 729$ and $a_2 + a_4 = \frac{111}{4}$, then $24\left(a_1 + a_2 + a_3\right)$ is equal to

A) 128 B) 129 C) 131 D) 130

52. Let $y = y(x)$ be the solution of the differential equation $\frac{dy}{dx} + 3\left(\tan^2 x\right)y + 3y = \sec^2 x, y(0) = \frac{1}{3} + e^3$. Then $y\left(\frac{\pi}{4}\right)$ is equal to

A) $\frac{4}{3}$ B) $\frac{2}{3} + e^3$ C) $\frac{4}{3} + e^3$ D) $\frac{2}{3}$

53. Bag A contains 2 white, 1 black and 3 red balls and bag B contains 3 black, 2 red and n white balls. One bag is chosen at random and 2 balls are drawn simultaneously from it, are found to be 1 red and 1 black. If the probability that both balls come from Bag A is $\frac{6}{11}$, then n is equal to

A) 13 B) 6 C) 4 D) 3

54. Let the product of roots of the equation $\sqrt{2026}(x)^{\log_{2026} x} = x^2$ is k^2, then k is equal to

A) 2025 B) 2026 C) 2027 D) 2028

55. Let $A\left(\frac{3}{\sqrt{a}}, \sqrt{a}\right)$ be a fixed point in xy-plane. The image of A in y-axis be B and the image of B in x-axis be C. If $D\left(3\cos\theta, a\sin\theta\right)$ is a point in the fourth quadrant such that the maximum area of ΔACD is 12 square units, then 'a' is equal to

A) 4 B) 5 C) 6 D) 8

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56. For $m,n>0$, let $\alpha(m,n)=\int_0^2 t^m(1+3t)^n\,dt$. If

$11\alpha(10,6)+18\alpha(11,5)=p(14)^6$, then '$p$' is equal to

A) 32 B) 24 C) 18 D) 16

57. For some positive integers 'k', the parabola with equation $y=\frac{x^2}{k}-5$ intersects the circle with equation $x^2+y^2=25$ at exactly three distinct points A,B and C. Sum of all positive integers 'k' for which the area of ΔABC is an integer, is

A) 22 B) 25 C) 16 D) 18

58. If the shortest distance between the lines

$\vec{r}=\left(-\hat{i}+3\hat{k}\right)+\lambda\left(\hat{i}-a\hat{j}\right)$ and $\vec{r}=\left(-\hat{j}+2\hat{k}\right)+\mu\left(\hat{i}-\hat{j}+\hat{k}\right)$ is $\sqrt{\frac{2}{3}}$, then the integral value of 'a' is equal to

A) 3 B) 4 C) 2 D) 6

59. Let $f:R\to R$ satisfy the equation $f(x+y)=f(x).f(y)$ for all $x,y\in R$ and $f(x)\neq 0$ for any $x\in R$. If the function f is differentiable at $x=0$ and $f'(0)=3$, then $\lim_{h\to 0}\frac{1}{h}\left(f(h)-1\right)$ is equal to

A) 2 B) 3 C) 4 D) 6

60. The total number of three-digit numbers with one digit repeated exactly two times is

A) 216 B) 225 C) 234 D) 243

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61. If $\left(\sin^{-1}x\right)^2-\left(\cos^{-1}x\right)^2=a,\ 0<x<1,\ a\neq 0$, then the value of $\left(2x^2-1\right)$ is

A) $\cos\left(\frac{4a}{\pi}\right)$ B) $\sin\left(\frac{2a}{\pi}\right)$ C) $\cos\left(\frac{2a}{\pi}\right)$ D) $\sin\left(\frac{4a}{\pi}\right)$

62. Let $f:[-2,3]\to[0,\infty)$ be a continuous function such that $f(1-x)=f(x)$ for all $x\in[-2,3]$. If R_1 is the numerical value of the area of region bounded by $y=f(x),\ x=-2,\ x=3$ and the axis of x and $R_2=\int_{-2}^{3}xf(x)dx$, then

A) $3R_1=2R_2$ B) $2R_1=3R_2$ C) $R_1=R_2$ D) $R_1=2R_2$

63. The parabola $y^2=x$ divides the circle $x^2+y^2=2$ into two parts whose areas are in the ratio

A) $9\pi+2:\ 3\pi-2$ B) $9\pi-2:3\pi+2$

C) $7\pi-2:2\pi-3$ D) $7\pi+2:3\pi+2$

64. If $\left(0,\ 3+\sqrt{5}\right)$ is a point on the ellipse whose foci are $(2,3)$ and $(-2,3)$, then the length of semi-major axis is

A) 6 B) 8 C) 3 D) 4

65. The sum of $\frac{7}{2\times3}\left(\frac{1}{3}\right)+\frac{9}{3\times4}\left(\frac{1}{3}\right)^2+\frac{11}{4\times5}\left(\frac{1}{3}\right)^3+....$ upto 10 terms is equal to

A) $\frac{1}{2}-\frac{1}{12\times3^{10}}$ B) $\frac{1}{3}-\frac{1}{12\times3^{10}}$ C) $\frac{1}{2}-\frac{1}{10\times3^{10}}$ D) $\frac{1}{3}-\frac{1}{10\times3^{10}}$

66. A coin is tossed three times. Let X denote the number of times a tail follows a head. If μ and σ^2 denote the mean and variance of X, then the value of $64\left(\mu^2+\sigma^2\right)$ is

A) 48 B) 64 C) 32 D) 51

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67. Let $A=\{0,3,4,6,7,8,9,10\}$ and R be the relation defined on A such that $R=\{(x,y)\in A\times A: x-y \text{ is odd positive integer or } x-y=2\}$. The minimum number of elements that must be added to the relation R, so that it is a symmetric relation is equal to

A) 15 B) 16 C) 17 D) 19

68. If the circle $C_1: x^2+y^2=16$ intersects another circle C_2 of radius 5 in such a manner that the common chord is of maximum length and has slope equal to $\frac{3}{4}$, then the co-ordinates of the centre of C_2 can be

A) $\left(\frac{9}{5},\frac{12}{5}\right)$ B) $\left(\frac{-5}{9},\frac{-4}{9}\right)$ C) $\left(\frac{9}{5},\frac{-12}{5}\right)$ D) $\left(\frac{-5}{9},\frac{4}{9}\right)$

69. Let $A=\{n\in N: \text{H.C.F.}(n,45)=1\}$ and $B=\{2k: k\in\{1,2,.....,100\}\}$, then the sum of all the elements of $A\cap B$ is

A) 5264 B) 4634 C) 5624 D) 4364

70. Let O be origin, the points $A(z_1)$ and $B(z_2)$ are such that $z_1=\sqrt{3}+2\sqrt{2}i$ satisfying $\sqrt{3}|z_2|=|z_1|$ and $\arg(z_2)=\arg(z_1)+\frac{\pi}{6}$, then

A) area of triangle ABO is $\frac{11}{4}$

B) Area of triangle ABO is $\frac{11}{\sqrt{3}}$

C) ABO is a scalene triangle

D) ABO is an obtuse angled isosceles triangle

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SECTION – II
(Numerical Value Answer Type)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only**. Have to **answer 5 questions** and question will be evaluated according to the following marking scheme.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

71. Let M be a 3×3 non-singular matrix with $\det(M)=4$. If $M^{-1}adj(adj\ M)=k^2I$, then the value of $|k|$ is _____

72. If $\sum_{r=0}^{2026} r^2\ {}^{2026}C_r = 2026\times\alpha\times2^{2024}$, then the number of positive prime factors of 'α' is

73. Let $\vec{a}=\hat{i}-\hat{j}+2\hat{k}$ and $\vec{b}$ be a vector such that $\vec{a}\times\vec{b}=2\hat{i}-\hat{k}$ and $\vec{a}.\vec{b}=3$ such that the projection of $\vec{b}$ on the vector $\vec{a}-\vec{b}$ is $\frac{\lambda}{\sqrt{\mu}}$; λ and μ are co-prime naturals, then μ is equal to _____

74. Area of the region bounded by the curve $y=\max\{|x|, x|x-2|\}$, x-axis and the lines $x=-2$ and $x=4$ is equal to _____

75. Let P be the foot of perpendicular from the point $Q(10,-3,-1)$ on the line $\frac{x-3}{7}=\frac{y-2}{-1}=\frac{z+1}{-2}$, such that the area of triangle PQR, where R is the point $(3,-2,1)$, is $a\sqrt{b}$; where $a,b\in N$, then the product 'ab' is equal to____

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